MIL-T-37512A
24 August 1978
SUPERSEDING
MIL-T-37512(DSA-DM)
8 March 1976

MILITARY SPECIFICATION

THEOPHYLLINE ELIXIR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Theophylline Elixir (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, Defense Personnel Support Center, ATTN: Directorate of Medical MAteriel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be an elixir containing the following ingredients:

Theophylline Anhydrous	0.533 percent
Alcohol	20.0 percent
Saccharin Sodium	0.1 percent
Flavors, Colors, Other vehicles, as appropriate	
Purified Water	dilute to volume (100.0 percent)

3.2 Assay.

3.2.1 Theophylline. The elixir shall contain not less than 76.0 mg and not more than 84.0 mg of Theophylline per 15.0 ml, calculated on an anhydrous basis (not less than 95.0 percent and not more than 105.0 percent of the required amount of Anhydrous Theophylline), when determined by the method described in 4.7.1.

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3.2.2 Alcohol. The elixir shall contain not less than 18.0 percent and not more than 22.0 percent of alcohol when determined using the USP Distillation Method.

3.2.3 Other ingredients. The content of sodium saccharin, purified water, flavoring and coloring agents, and other vehicles in the elixir shall be determined from the batch formulation sheet kept on file by the manufacturer. The content of the sodium saccharin shall be in conformance with the formulation specified in 3.1. No ingredient, such as sugar, shall be used in the formulation that is contraindicated for use by diabetic patients.

3.3 Palatability. The finished elixir shall be palatable and pleasant to the taste with no unpleasant after-taste. Not later than the time specified for opening the bids or receipt of the proposals, the offeror shall submit to the contracting officer six 4 fl. oz. bottles of elixir which are representative of the product which the offeror proposes to furnish. A portion of the sample will be subjected to panel testing for a determination of palatability (see 4.7.2 Palatability test). A portion will be used by the cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the specification requirement.

3.4 Identification of theophylline in the finished elixir.

3.4.1 Melting range. The precipitate obtained when the elixir is treated as specified in 4.7.3.1 shall melt between 270° and 274°C when determined by the USP method.

3.4.2 Color test. The precipitate obtained in 4.7.3.1 shall develop a purple color when treated as described in 4.7.3.2 which is destroyed by solutions of fixed alkalis.

3.5 Ingredients.

3.5.1 Theophylline and alcohol. The theophylline and the alcohol used in the manufacture of the elixir shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

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3.5.2 Other ingredients. All other ingredients used in the manufacture of the elixir shall be of USP or NF quality, or if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.6 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the elixir to the date of delivery to the Government.

3.7 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item. for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of elixir produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. For raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Theophylline Assay	3.2.1	S-2	1
Alcohol Assay	3.2.2	S-2	1
Other Ingredients	3.2.3	-	-
Palatability	3.3	-	-
Theophylline Identification	3.4	S-2	1

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished elixir shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

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TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Elixir not free from foreign matter.
102	Elixir has crystal formation.
103	Elixir not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Theophylline assay. Accurately weigh approximately 100 mg of theophylline USP reference standard into a 100 ml volumetric flask, add 50 ml of purified water and 1 ml of hydrochloric acid. Shake until solution is complete and dilute to volume with purified water. Pipet 1.0 ml of the solution into a 100 ml volumetric flask, add 50 ml of purified water, 1 ml of hydrochloric acid, and dilute to volume with purified water.

Pipet 1.0 ml of elixir into a 500 ml volumetric flask, add 400 ml of purified water, 5 ml of hydrochloric acid, and dilute to volume with purified water. Concomitantly determine the absorbances of standard and sample solutions in 1 cm cells, with a suitable spectrophotometer at the peak absorbance at 270 ± 2 nm, using a solution of *1 ml hydrochloric acid diluted to a 100 ml with purified water as the blank.*

Calculation: $\frac{Au}{As} \times \frac{\text{mg standard}}{100} \times \frac{1}{100} \times \frac{500}{1} \times \frac{15}{1} = \text{mg theophylline per 15 ml}$

$$\frac{\text{mg theophylline}}{80} \times 100 = \text{percent required amount theophylline}$$

where: Au = absorbance of sample
As = absorbance of standard

As an alternate assay procedure, the NF method for Theophylline, Ephedrine Hydrochloride, and Phenobarbital Tablets shall be employed with the following modifications:

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At the bottom of a chromatographic tube approximately 18 mm (I.D.) x 230 mm in length with a 19/22 fitting, add a 12 cm delivery tube with sintered glass perforated plate (*Scientific Glass Catalogue No. JC2100 Size II, or* an equivalent tube). Pack a pledget of fine glass wool in the base of the column. To 500 mg of purified, acid washed siliceous earth, mesh size 45-60, add 0.3 ml of 0.67 tribasic sodium phosphate, and incorporate by kneading to a uniform mixture. Transfer above to the chromatographic column and pack lightly with a tamping rod. Transfer 10.0 ml of the finished elixir to a beaker. Add 2.0 ml of 0.67 M tribasic sodium phosphate and mix well with the sample. Add 9 grams of the siliceous earth, mix thoroughly and transfer to the column in several portions packing each portion uniformly. Place a pledget of fine glass wool on top of the column and pack firmly.

Wash the column with three 25 ml portions of water-washed chloroform, allowing each portion to pass through the column before adding the next portion. Discard the washings.

Elute the column in accordance with the NF procedure for column A, and shake the eluate with dilute sulfuric acid solution as directed. Discard the chloroform solution. Treat the aqueous solution as directed in the NF monograph.

Prepare a Theophylline Standard solution as directed in the monograph. Concomitantly determine the absorbances of the standard and sample solutions with a suitable spectrophotometer, in a 1 cm cell, at the maximum absorption at about 275 nm, versus 0.5 N sodium hydroxide solution in the reference cell. Calculate the concentration of Theophylline in the finished elixir.

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4.7.2 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the item's Palatability Standard 1/, when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the samples shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ The item's Palatability Standard is available, upon separate request to the contracting officer, Defense Personnel Support Center, Directorate of Medical Materiel.

4.7.3 Identification.

4.7.3.1 Melting range. Reduce the volume of about 25 ml of elixir to approximately 15 ml by heating on a steam bath. Add 10 ml of purified water and cool the solution in a refrigerator to approximately 10°C. and add 1 ml of hydrochloric acid. Collect the precipitate on a sintered glass filter of medium porosity. Wash the precipitate with several portions of cold purified water and dry for 1 hour at 105° C. Determine the melting point of the precipitate.

4.7.3.1.1 As an alternate omit the addition of the 1 ml of hydrochloric acid but follow the procedure in all other respects.

4.7.3.2 Transfer about 10 mg of the dried precipitate from 4.7.3.1 to a small porcelain evaporating dish. Add 1 ml of hydrochloric acid, 100 mg of potassium chlorate, and evaporate to dryness on a steam bath. Invert the dish over a vessel containing a few drops of Ammonia TS. Observe the color change of the precipitate.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 , and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 1 gal (3.78 liters) as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2 Unit packaging. Each unit shall be packaged as specified in 5.2.10 of PPP-C-186 in lieu of the intermediate package.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Four (4) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Four (4) units of issue shall be packed as specified in 5.4.3.4 of PPP-C-186.

* 5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

* 5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

* NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

* 5.4.2 Exterior container. Each exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legends:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)"
AND
"DO NOT PERMIT TO FREEZE"

* 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-781-8474	THEOPHYLLINE ELIXIR, 1 gal (3.78 liters)

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6.3 *Changes from previous issue*. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Governemnt assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-37512A THEOPHYLLINE ELIXIR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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